

SHIVANJALI VIJAYKUMAR RANASHING

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EDUCATION

PhD in Graph Mining and Machine Learning

Dec 2025

University of Cincinnati College of Engineering and Applied Science

- Researching algorithms for community detection in directed and signed graphs.
- Submitted a paper to the PAKDD and IJCAI conferences and am currently preparing submissions for the KAIS journal.
- **Relevant Courses:** Machine Learning, Deep Learning, Natural Language Processing, Intelligent Systems, Big Data Analytics, Complex Systems, Statistics and Machine Learning.

MS in Computer Science

New Jersey Institute of Technology

BE in Computer Engineering

Maharashtra Institute of Technology Pune [Savitribai Phule Pune University, India]

Diploma in Computer Engineering

Government Polytechnic Pune, India

SKILLS

- | | | |
|-------------------------------|------------------------|--------------------------------|
| • Python, Pytorch, TensorFlow | • Networkx, Neo4j | • Poetry |
| • LangChain | • FastAPI | • Github, Jenkins, AWS |
| • FastAPI | • Hugging Face, OpenAI | • Collaboration and Leadership |

PROFESSIONAL EXPERIENCE

Graduate Research Assistant at UC Sim Center, in collaboration with P&G

Oct 2020 to Present

- Utilized LangChain for prompt engineering and graph mining, employing OpenAI embeddings for exploratory analysis and graph searches, with Networkx for graph operations and ConvexHull for cluster analysis.
- Implemented spectral clustering to group similar data points and perform similarity analysis for extracting meaningful patterns and insights.
- Developed a pipeline to convert unstructured PDF content into graphs using Neo4j and implemented GraphRAG, leveraging OpenAI embeddings with vector indexing for information retrieval and Chat API for generating responses.

Summer Intern at Fidelity, Covington, Kentucky

June 2024 to Aug 2024

- Developed a Retrieval-Augmented Generation (RAG) workflow to generate summarized answers with references from PDF files, leveraging AWS Textract for content extraction and parsing JSON objects.
- Managed file operations with AWS S3 using boto3 and S3FS for efficient data handling and utilized sentence-transformers for embedding storage in vectors for content retrieval.
- Built a Python backend with FastAPI, integrated with a Streamlit frontend, and employed Hugging Face's summarization pipeline to distill retrieved content. Set up the project environment with Poetry and utilized AWS RDS for structured data storage.

Technology Associate at Morgan Stanley, New York

August 2017 to May 2020

Corporate & Funding Technology division, Prime Brokerage Tech department

- Experienced full-stack developer skilled in building enterprise-grade applications with a focus on financial services, utilizing Angular for front-end and Java with Spring, JPA, and Hibernate for backend development.
- Developed of Order Management and Anti-Money Laundering modules, working extensively with RESTful services, database management and deployed applications in Agile environments, leveraging Git, Jenkins for version control and deployment.

Summer, Fall and Spring Intern at ADP, Roseland, New Jersey

June 2016 to May 2017

- Developed a reporting application to retrieve and display historical data with graphical analysis, using Oracle Express and managing database operations with DDL and DML. Built and optimized a full-stack Netcool reporting tool with a Panorama VDL frontend, leveraging Angular for the UI, PHP for server-side data retrieval, and Tomcat for middleware, while implementing database indexing and partitioning for improved performance.

Software Engineer at ATOS, Pune, India, Nokia Project

- Rebranded Nokia Electronic Document and IB installers, managed hot deployment of WAR files on Tomcat, and delivered production enhancements using JSP, Servlet, Java, HTML, and InstallShield.

Virtual Money Sponsored Project by Persistent Systems Ltd.

- Developed an Android application for virtual point transactions, with a .NET C# server handling account management via SOAP-based XML communication.

PROJECTS

Image Analysis

March 2023

- Developed a color segmentation solution using the K-means algorithm to extract and list colors from images along with their respective percentages. Implemented image classification through transfer learning techniques.
- Built a color pigmentation data mining application utilizing the Orange Data Mining platform.

Chatbot using RAG framework

Oct 2024

- Using LangChain, ChromaDB, and the Mistral model, develop a chatbot that employs RAG (Retrieval-Augmented Generation) to provide accurate answers by retrieving relevant information from documents processed with PyPDF and embedded using FastEmbed. Deploy this chatbot with Streamlit and Streamlit-Chat for an interactive prototype.

SmartGraphBot

Oct 2024

- Leverages LangChain, LangChain Community, LangChain Experimental, LangChain Ollama, and LangChain Core to provide intelligent, context-aware responses by combining RAG (Retrieval-Augmented Generation) and knowledge graph insights from Neo4j.
- With a knowledge graph backend, the bot can understand complex relationships and retrieve accurate information in real time.
- Deployed on Gradio, the chatbot offers an intuitive interface for exploring relational data and deepening interactions.

TrueView

Nov 2024

- Without a dislike count, viewers may struggle to judge content quality. TrueView bridges this gap by inferring the general sentiment from comments, giving viewers a clear idea of the community's reaction. Chrome extension designed to provide viewers with instant insights into YouTube video sentiments.
- Developed a Chrome extension using JavaScript, FastAPI, and YouTube API to analyze user sentiment on YouTube videos via comments, leveraging LangChain for NLP processing. Enabled users to assess video quality and engagement value by identifying sentiment trends, compensating for the disabled dislike button.

ACCOMPLISHMENTS

- **Won Best Security Hack and Most Creative Use of GitHub** prizes at MakeUC Hackathon Nov 2023
 - <https://devpost.com/software/safe-roads>
- **Best Educational Hack Issued by RevolutionUC.** Associated with 1819 Innovation Hub Feb 2023
 - <https://devpost.com/software/amigo-62adsg>
- **Spark Grant Winner** powered by Fealy Bearcat Bridge Fund to discover entrepreneurs 2022
- **Google Cloud Platform Fundamentals: Core Infrastructure** issued by Coursera 2019
- **Oracle Certified Java SE 6 Programmer** Feb 2014
- **Achiever's Award** at ATOS Nov 2014
- **Certificate of Excellence with accolade silver** at ATOS Dec 2013
- **First prize in Blind Coding inter-college competition** [Intechxication] held at MIT, Pune March 2010

EXTRENA ACTIVITIES

- **Student Assistant for the Women's Outreach Program at NJIT (Apr 2016 - Dec 2016)**, organizing school visits and presentations to encourage high school and middle school girls to pursue careers in computer science.
- **BRAID (Building Recruiting and Inclusion for Diversity) Ambassador (2016)**, an initiative by the Anita Borg Institute, promoting diversity and inclusion in STEM, especially computing.
- **Elected Webmaster for the Graduate Women in Computing Science E-board at NJIT (2016)**, supporting digital communication and outreach efforts.
- **Received sponsorship from NJIT and the University of Cincinnati** to attend the prestigious Grace Hopper Celebration (2016) and Richa Tapia (2016) conferences.